

## Multiple Choice (10 marks)

1. Calculate the heat transferred when 250 J of work is done on a system consisting of 1 mole of an ideal gas. At constant temperature,  $\Delta U = 0$  for the expansion of an ideal gas.
  - (a) 1 J
  - (b) 500 J
  - (c) -250 J
  - (d) -500 J
  - (e) -100 J
2. Under which conditions will a real gas most closely behave as an ideal gas?
  - (a) Low temperature and low pressure
  - (b) Low temperature and high pressure
  - (c) High volume and high pressure
  - (d) High temperature and high pressure
  - (e) High volume and low pressure
3. What is the value of  $\log 0.0148$ ?
  - (a) 3.0000
  - (b) -0.1703
  - (c) -2.1703
  - (d) -0.8297
  - (e) -1.1703
4. Which of the following is classified as a conjugate acid-base pair?
  - (a)  $\text{HCl} / \text{NaOH}$
  - (b)  $\text{H}^+ / \text{Cl}^-$
  - (c)  $\text{H}_3\text{O}^+ / \text{H}_2\text{O}$
  - (d)  $\text{O}_2 / \text{H}_2\text{O}$
5. How many ounces of silver alloy which is 28% silver must be mixed with 24 ounces of silver which is 8% silver to produce a new alloy which is 20% silver?
  - (a) 48 ounces of silver
  - (b) 42 ounces of silver
  - (c) 54 ounces of silver
  - (d) 72 ounces of silver
  - (e) 60 ounces of silver

## True / False (10 marks)

*Answer True or False*

6. True or False: The reaction proceeds with 100% efficiency, producing only n-butane without any side reactions.
7. True or False: para-dichlorobenzene can be correctly referred to as 2,4-dichlorobenzene, which indicates a different substitution pattern.
8. True or False: The Boyle temperature of a van der Waals gas can be expressed as  $T_B = [(a+b) / (R)]$ .
9. True or False: The energy of a photon associated with the series limit of the Lyman series is 123.64 nm.
10. True or False: A silver dollar weighs approximately 26.73 grams, which is equivalent to 0.02673 kilograms and 26,730 milligrams.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Calculate the volume percentage of an E. coli culture at its limiting concentration of  $10^9$  cells per  $cm^3$ , given the dimensions of a cylindrical cell are 2.0  $\mu m$  in length and 1.0  $\mu m$  in diameter. Discuss your process and findings.
12. Discuss the net ionic equation that results from the reaction between  $NH_4Br$  and  $AgNO_3$  in aqueous solutions, including the significance of the ions involved and the overall reaction process.

*End of Paper*